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A neural network-based selective harmonic elimination scheme for five-level inverter

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Abstract

This paper presents the implementation of selective harmonic elimination (*SHE*) in a five-level inverter structure using artificial neural networks (*ANNs*). *SHE* is an effective low-frequency modulation technique to eliminate selected harmonics and control multilevel converters. The use of *ANN-SHE* requires the calculation of the optimum values of switching angles via the solving system of nonlinear equations for the total harmonic distortion (*THD*) reduction, where the nonlinear equations are founded by the complex Fourier series analysis of the inverter output voltage. The procured switching angle values are directly implemented by a multilayer perceptron (*MLP*) algorithm without a lookup table. The *ANN* model is obtained by training the neural network (*NN*), taking the modulation index (*M*) as an input and approximating switching angles as an output. A thorough analysis was carried out to show the programming steps of the proposed ANN-based *SHE* using Matlab/Simulink environment. A realized inverter prototype steered by the proposed ANN-based *SHE* was tested with various modulation indexes on a real-time mode using a digital signal processor (*DSP*) C2000 Delfino–TMS320F28379D-embedded board. A comparison between the simulation results and the experimental data is presented. The obtained results illustrate that the experimental results match the simulation closely, and the *ANN* model provides a fast and precise estimate of the switching angles for each value of the modulation index.

KEYWORDS

artificial neural networks (*ANNs*), C2000 Delfino (*DSP*) microcontroller, five-level inverter, power electronics, selective harmonic elimination (*SHE*)

Abbreviations: ω_s, switching frequency; *AC*, alternating current; *AI*, artificial intelligence; *ANN*, artificial neural network; *CHB*, cascaded half-bridge; *D*, diode; *DC*, direct current; *DGS*, distributed generation systems; *DSP*, digital signal processor; *f*, frequency; *FFT*, fast Fourier transform; *FPGA*, field programmable gate array; *GPS*, generalized pattern search; *GSA*, gravitational search algorithm; *GWO*, grey wolf optimizer; *Hz*, hertz; i_o, output current; *M*, modulation index; *MLI*, multilevel inverter; *MLP*, multilayer perceptron; *MPPT*, maximum power point tracking; *N-R*, Newton–Raphson algorithm; *PV*, photovoltaic; *S*, switch; *SHE*, selective harmonic elimination; *SHEPWM*, selective harmonic elimination pulse width modulation; *T*, switching period; *THD*, total harmonic distortion; *THDv*, total harmonic distortion of voltage; U_o, output voltage; *V*, voltage; V_{dc}, direct current voltage; θ, switching angle.

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1 | INTRODUCTION

In recent decades, several emerging multilevel inverters (*MLIs*) have been proposed for many industrial applications and pedagogical studies to overcome conventional inverters problems. The proposed new converters feature fewer isolated *DC* power supplies, fewer switches, and fewer gate drivers requirement and generate high-power quality with low total standing voltage.[1–4] For an efficient operation of a *MLI*, several modulation techniques are suggested for their control. Modulation strategies for *MLIs* are defined as techniques or methodologies for producing pulses for turning on and turning off the power semiconductor devices. There are currently three modulation modes based on switching frequency: low-frequency switching modulation, high-frequency switching modulation, and mixed-frequency switching modulation.[5,6] The selective harmonic elimination (*SHE*) technique offers several advantages when it is used to generate the gating signals of the *MLI*, where the switching angles are determined by solving a nonlinear transcendental system of equations. In literature, several methods have been proposed to find the best switching angles,[7] including Newton–Raphson (*N-R*) algorithm.[8] The *N-R* method necessitates an initialization of the switching angles near to the optimum solution in order to converge correctly. It is challenging task to estimate the initial solutions of a system of nonlinear equations, particularly for a huge number of switching angles. On the other hand, several optimization procedures were developed in order to find the optimal switching angles using optimization algorithms, in which the *SHE* equations are presented in a cost function that can be optimized. Siddique et al.[9] presented a *SHE* pulse width modulation (*SHEPWM*) technique for an improved single-phase inverter. The performance and effectiveness of the proposed topology have been verified by the obtained simulation and experimental results with the *SHEPWM* control. Haghdar et al.[10] proposed a generalized pattern search (*GPS*) algorithm for the harmonic elimination in five-level and seven-level inverters. K. P. Panda and G. Panda[11] have presented a particle swarm optimization (*PSO*) based on a modified *SHE* technique to find optimum switching angles of the proposed 6S-7L *MLI*. The performance of the proposed *PSO-SHE* was verified by the simulation and experimental results for eliminating the third- and fifth-order harmonics. Ganesan et al.[12] proposed *SHE* of cascaded seven-level *MLI* using the bat evolutionary optimization method. K. P. Panda et al.[13] proposed *SHE* for developed asymmetrical *MLI*, where the optimum switching angle is computed using the new flower pollination algorithm (*FPA*). The simulation and experimental results demonstrated the performance of the presented *FPA-SHE*. Lodi et al.[14] have presented a *SHE* for a modified packed U-cell five-level inverter, where the optimum switching angle is computed using genetic algorithm (*GA*). The implementation of *SHE* has grabbed the attention of researchers for robust and effective applications. Nevertheless, the selection of switching angles to eliminate a specific harmonic is one of the *SHE's* ongoing research challenges. There has always been a need for a solid knowledge of mathematical analysis and programming techniques for solving nonlinear equations to get the desired switching angles.

In the artificial intelligence (*AI*) field, artificial neural network (*ANN*) proved its effectiveness in several applications such as function approximation, trajectory optimization, pattern recognition, nonlinear systems control, faults detection, and classification.[15–17] Kermadi et al.[18] proposed the study of AI-based maximum power point tracking controllers (*MPPTs*) for photovoltaic (*PV*) systems. Vinnakoti et al.[19] have presented a three-level converter-based unified power quality conditioner (*UPQC*) using ANN-based control scheme. Mohamed et al.[20] have proposed a neural network (*NN*) for model predictive control (*MPC*) of a three-phase inverter. Namigtle-Jiménez et al.[21] have presented an online ANN-based fault diagnosis implementation using a field programmable gate array (*FPGA*).

The present work proposes an ANN-based *SHE* technique to control a five-level inverter; the objective is to solve the equations of *SHE* and implement the ANN-based controller in real-time using a low-cost embedded board digital signal processor (*DSP*) C2000 Delfino–TMS320F28379D. The *ANN* is trained to map the inputs (modulation index) and outputs (optimum switching angles) of the *SHE* control. The proposed *ANN-SHE* eliminates the most unwanted harmonics, i.e., third harmonic and its multiple, responsible for over-currents generation and overheating equipment. The paper's organization is as follows: the second section presents the proposed hybrid five-level inverter, i.e., the inverter structure and its principle of operation. Next, the third section discusses the *SHE* technique and principle of NN-based *SHE*. Then, the fourth section presents the results of the simulation and experiment with comparison of the *ANN-SHE* with the *N-R* and *GA* algorithms. A comparison with the high-frequency modulation technique is also provided. The fifth section is devoted to give the most critical applications of the five-level inverter with *ANN-SHE*. Finally, the sixth section provides the conclusions of the study.

2 | ANALYSIS OF THE FIVE-LEVEL INVERTER OPERATION

Figure 1 illustrates the block diagram of the *ANN-SHE* control scheme, and the hybrid proposed five-level inverter topology, which contains six MOSFET switches and two separate *DC* sources. This structure is made up of a reduced number of switches which is compared with that of the conventional topology of the five-level cascaded H-bridge inverter.[22] The inverter can produce five different voltage levels: $+V_{dc}$, $+2V_{dc}$, 0, $-V_{dc}$, and $-2V_{dc}$.

Table 1 shows the switching modes of topology that were proposed in this work. The load output voltage ($V_o = U_{ab}$) comprised five voltage states in an entire switching period (T). Additionally, there are 10 switching states of the five-level inverter in the entire switching period (T) since the load current is bi-directional. The pair switches (S_1 and S_2), (S_3 and S_4), and (S_5 and S_6) require to be operated in a complementary mode with dead time in order to not to have a voltage source short circuit as well as having a continuous conducting with a complementary mode of control must be defined. Figure 2 shows the equivalent circuits for generating five-level voltage, and Figure 3 shows the switching pattern and the desired voltage waveform of the five-level inverter for a complete cycle.

The output voltage of the inverter $U_{ab}(t)$ could be regulated by modulating the switching frequency ω_s or controlling the angle switching, θ_1 and/or θ_2, so the operating point was determined by (V, ω_s, θ_1, θ_2).

This work aims to get rid of the third-order (3rd) harmonics, so when Fourier analysis is applied, the voltage $U_{ab}(t)$ can be created with the following Equations 1–4:

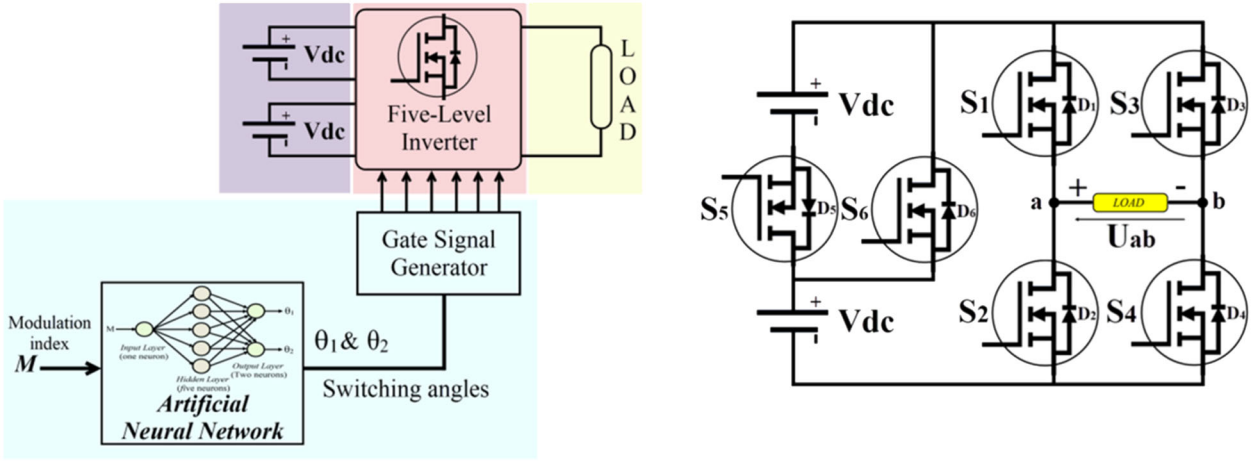


FIGURE 1 Block diagram of *ANN-SHE* control scheme and the five-level inverter structure [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Switching states of the five-level topology

Interval	V_o	ON switches	OFF switches	Positive current $i_o(+)$ path	Negative current $i_o(-)$ path
$[0, \theta_1]$	0	S_2, S_4	S_1, S_3, S_5, S_6	S_4, D_2	D_4, S_2
$[\theta_1, \theta_2]$	$+V_{dc}$	S_1, S_4, S_6	S_2, S_3, S_5	S_4, D_6, S_1	D_4, S_6, D_1
$[\theta_2, \pi/2]$	$+2V_{dc}$	S_1, S_4, S_5	S_2, S_3, S_6	S_4, S_5, S_1	D_4, D_5, D_1
$[\pi/2, \theta_3]$	$+2V_{dc}$	S_1, S_4, S_5	S_2, S_3, S_6	S_4, S_5, S_1	D_4, D_5, D_1
$[\theta_3, \theta_4]$	$+V_{dc}$	S_1, S_4, S_6	S_2, S_3, S_5	S_4, D_6, S_1	D_4, S_6, D_1
$[\theta_4, \theta_5]$	0	S_2, S_4	$S_1, S_3, S_5 S_6$	S_4, D_2	D_4, S_2
$[\theta_5, \theta_6]$	$-V_{dc}$	S_2, S_3, S_6	S_1, S_4, S_5	S_3, D_6, S_2	D_3, S_6, D_2
$[\theta_6, 3\pi/2]$	$-2V_{dc}$	S_2, S_3, S_5	S_1, S_4, S_6	S_3, S_5, S_2	D_3, D_5, D_2
$[3\pi/2, \theta_7]$	$-2V_{dc}$	S_2, S_3, S_5	S_1, S_4, S_6	S_3, S_5, S_2	D_3, D_5, D_2
$[\theta_7, \theta_8]$	$-V_{dc}$	S_2, S_3, S_6	S_1, S_4, S_5	S_3, D_6, S_2	D_3, S_6, D_2
$[\theta_8, 2\pi]$	0	S_1, S_3	$S_2, S_4, S_5 S_6$	S_3, D_1	D_3, S_1

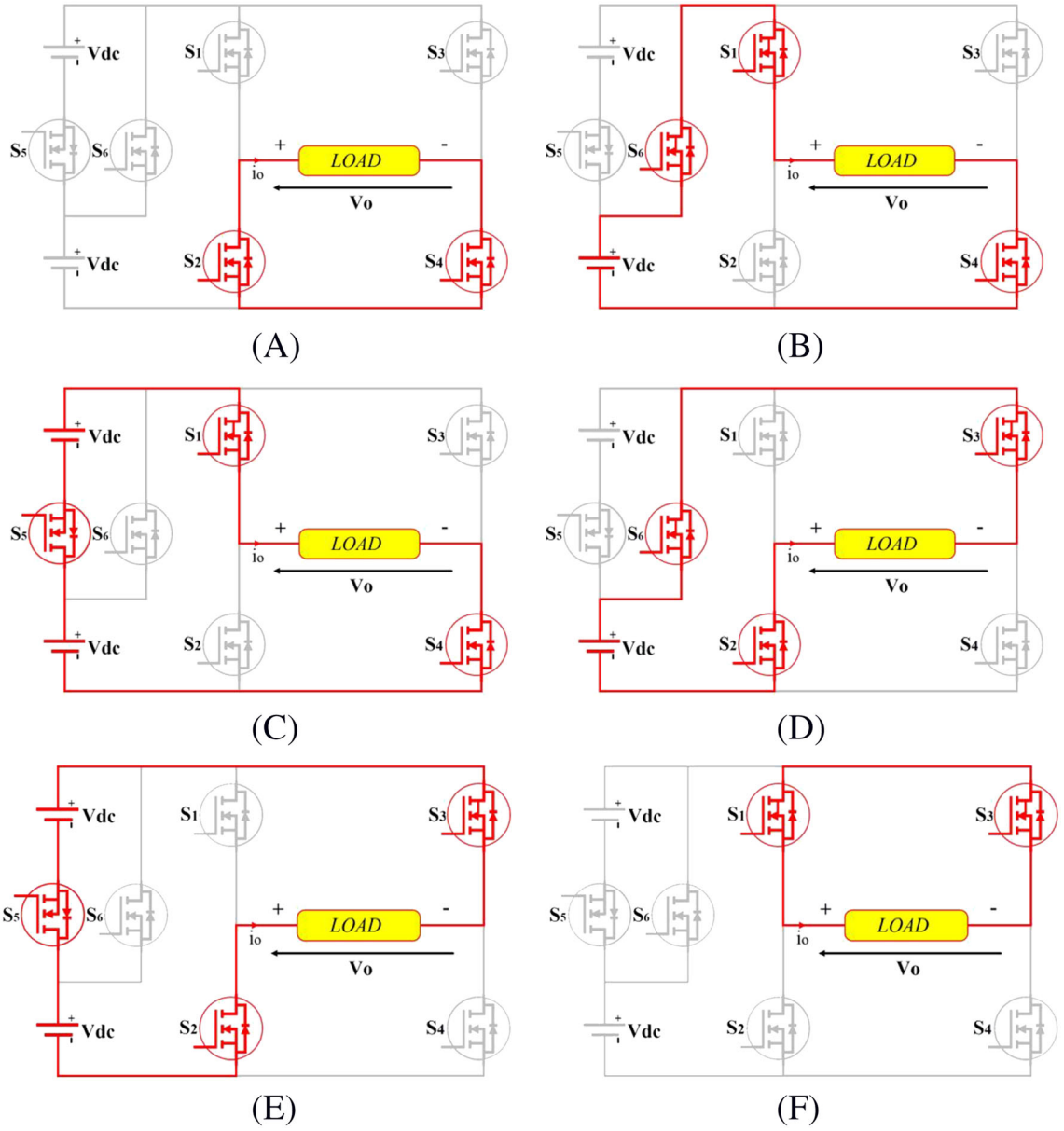


FIGURE 2 The switching states of the five-level inverter (A) $V_o = 0$, (B) $V_o = +V_{dc}$, (C) $V_o = +2V_{dc}$, (D) $V_o = -V_{dc}$, (E) $V_o = -2V_{dc}$, and (F) $V_o = 0$ [Colour figure can be viewed at wileyonlinelibrary.com]

$$U_{ab}(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_s t) + \sum_{n=1}^{\infty} b_n \sin(n\omega_s t) \tag{1}$$

where:

$$a_0 = 0,\ a_n = 0,\ b_n = \frac{2}{T}\int_0^T U_{ab} \sin(n\omega_s t)dt \tag{2}$$

$$b_n = \frac{4V_{dc}}{n\pi}(\cos(n\theta_1) + \cos(n\theta_2)) \tag{3}$$

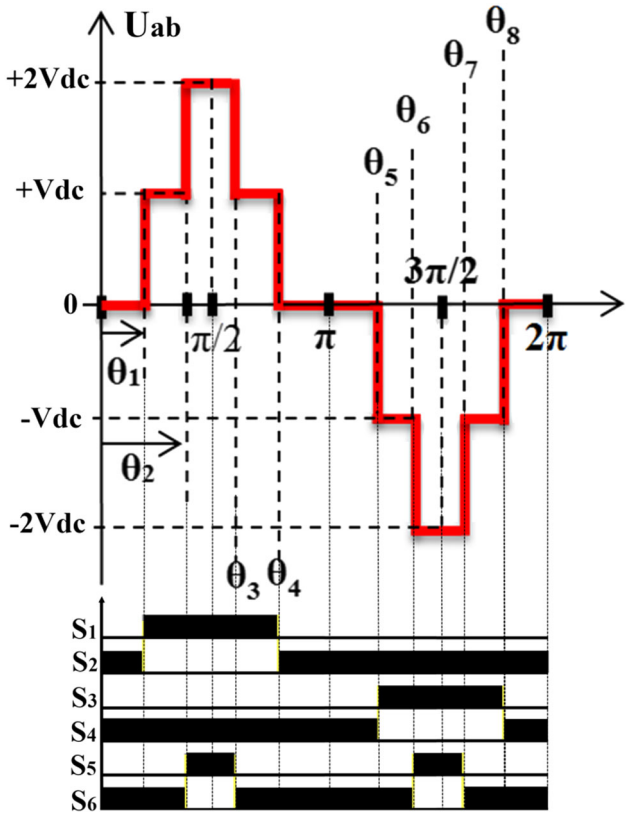


FIGURE 3 The switching pattern and desired five-level voltage: Half-waveform [Colour figure can be viewed at wileyonlinelibrary.com]

$$U_{ab}(t) = \sum_{n=1}^{\infty} b_n \sin(n\omega_s t) = \sum_{n=1}^{\infty} \left(\frac{4V_{dc}}{n\pi} (\cos(n\theta_1) + \cos(n\theta_2)) \right) \sin(n\omega_s t) \tag{4}$$

with the first harmonic ($n = 1$), the output voltage $U_{ab}(t)$ can be indicated as (5):

$$U_{ab}(t) = \frac{4V_{dc}}{\pi} (\cos(\theta_1) + \cos(\theta_2)) \sin(\omega_s t) \tag{5}$$

To eliminate the third harmonic and their multiple, we must search the switching angles θ_1 and θ_2 versus the modulation index M. The switching angles must satisfy the subsequent algebraic system of the nonlinear equations:

$$\begin{cases} (\cos(\theta_1) + \cos(\theta_2)) & = \dfrac{\pi h_1}{4V_{dc}} \\ (\cos(3\theta_1) + \cos(3\theta_2)) & = 0 \end{cases} \tag{6}$$

3 | THE ANN-BASED *SHE* TECHNIQUE

The neuron model is an elementary calculation unit that calculates a single output from the received information inputs or inputs. The inputs are weighted by W and an activation function to calculate the output.[15] The following mathematical expressions 7 and 8 govern the neuron behavior:

$$S = X_1W_1 + X_2W_2 + X_3W_3 + ... + X_nW_n = \sum_{i=1}^{n} X_iW_i \tag{7}$$

$$Y = f(s) \tag{8}$$

where X_1, X_2... X_n are the inputs of the artificial neuron, W_1, W_2... W_n are the synoptic weights of the artificial neuron, S is the sum, Y is outputs of the artificial neuron, and f is the transfer functions.

An *ANN* is a combination of connected artificial neurons; several architectures of *ANN* have been proposed. The multilayer perceptron (*MLP*) network type is organized into several layers in which information circulates starting the input to the output layers; it is a feed-forward *NN*. Each layer is made up of a variable number of neurons with the neurons in the last layer present the outputs of the system.[15–23] In this study, we used an *MLP* network with (1:5:2) structure as shown in Figure 4. (1:5:2) structure comprises an input layer that receives data (modulation index *M*), a hidden layer composed of five neurons and an output layer formed of two neurons (switching angles θ_1 and θ_2).

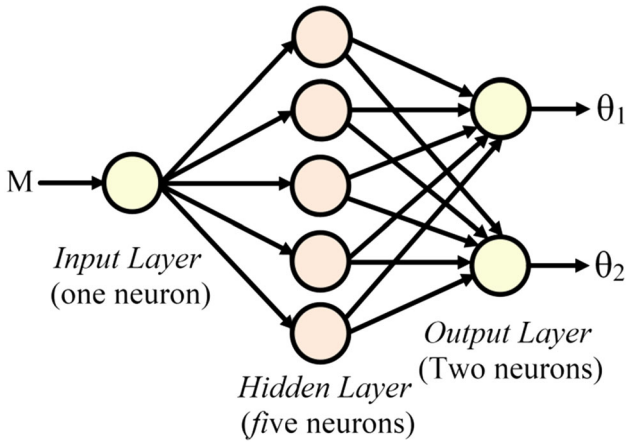


FIGURE 4 *MLP* neural network (1:5:2) structure [Colour figure can be viewed at wileyonlinelibrary.com]

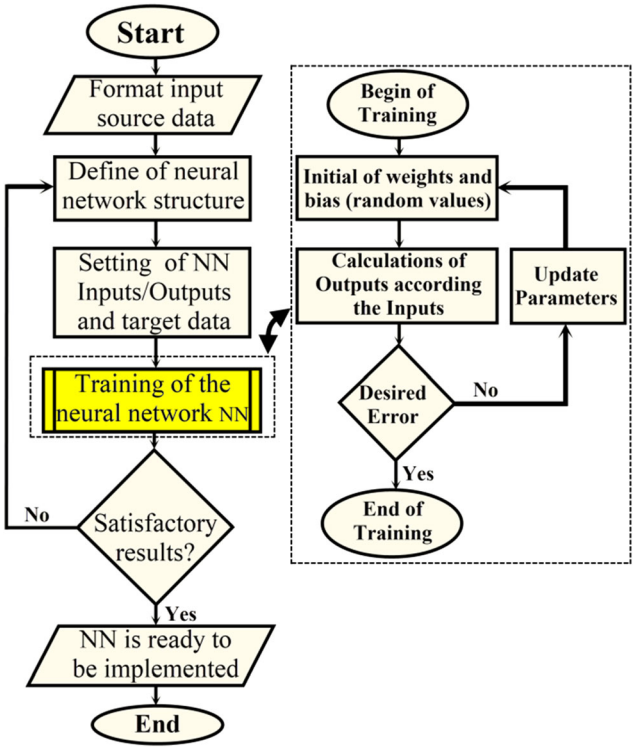


FIGURE 5 Flowchart of the *ANN-SHE* technique [Colour figure can be viewed at wileyonlinelibrary.com]

Figure 5 shows the flow chart of training *ANN-SHE*. The use of the backpropagation algorithm in the training of the feed-forward *NN* for supervised learning is done. With this method, the *NN* is updated until the desired objective is attended. The weights of connections are adjusted several times before archiving the final values. Figure 6 shows the proposed pseudocode to determine the switching angles θ_1 and θ_2 using the ANN-based *SHE* method. Figure 7 shows the gating switching angles θ_1 and θ_2 generated using the *ANN* to get rid of the third harmonic components, considering the modulation index M values. Following the analytical analysis,[24] the valid interval of modulation index is $M \in [0.45; 0.85]$ to eliminate the third harmonic and their multiple.

```
clc; clear all;
[a,b,c]=textread('ANN-SHE_DATA.txt','%f %f %f');
x=a';
y=[b c]';
% Neural Network structure
inputs  = [1]' % Input vector size
outputs = [1 2]' % Output vector size
net = network( 1,2,[1; 0],[1; 0],[0 0; 1 0], [0 1]);
net.layers[2].size = 5; % Number of neurons in hidden
layer
net.layers[2].transferFcn = 'logsig';% hidden layer transfer
function
net = configure(net,inputs,outputs);
view(net);
% Neural Network Training
net.trainFcn = 'trainlm';
net.performFcn = 'mse';
net = train(net,inputs,outputs);
[net,tr]  = train(net,x,y);
gensim(net,1)
% For plot the switching angles with ANN-SHE control
k=0.01:0.01:1;
net(k);
plot(k,net(k));
axis([0.4,0.9,0,100])
grid
xlabel('Modulation index M');
ylabel('Switching Angles(°  )');
legend('\theta1','\theta2');
```

FIGURE 6 Matlab pseudocode for ANN-based *SHE* control [Colour figure can be viewed at wileyonlinelibrary.com]

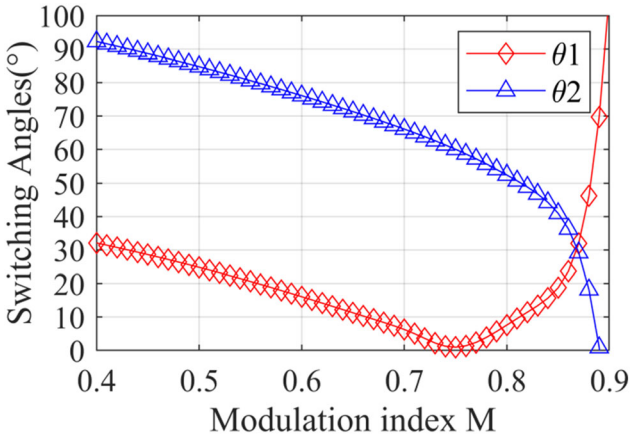


FIGURE 7 The switching angles θ_1 and θ_2 with *ANN-SHE* control [Colour figure can be viewed at wileyonlinelibrary.com]

Figure 8 shows the convergence of the mean square error (*MSE*) of the *NN* during training, testing and validating. As shown, the *MSE* for training was found at 0.016, the *MSE* for testing was found at 0.13, and the *MSE* for validating was found at 0.023 after 118 epochs. The *ANN* converged fast within the first epochs.

4 | RESULTS AND DISCUSSIONS

This section discusses the results of simulation and experiment, after that a comparison of obtained results with other modulation techniques and other algorithms. The proposed *ANN-SHE* control of the five-level inverter is tested with dynamic variation in modulation index *M*, where four scenarios are selected as indicates in Table 2. The operating parameters of system are $f = 50$ Hz and $V_{dc} = 27.5$ V.

4.1 | Simulation results

The simulation model of the single-phase five-level inverter with *ANN-SHE* control is built under a Matlab/Simulink environment. Figure 9A shows the five-level output voltage without *SHE* control under scenario 1. Their associated harmonic spectrums are presented in Figure 9B, where all harmonics exist. The total harmonic distortion (*THD*) voltage possesses a value of ($THDv = 30.21\%$). Afterward, a variation of the values was carried out on the switching angles, i.e., variation of modulation index *M*. Figure 10A shows the five-level output voltage with *ANN-SHE* control under Scenario 2. Its associated harmonic spectrum is presented in Figure 10B. The *THD* voltage possesses a value of ($THDv = 29.69\%$) and the third harmonic with their multiples have been eliminated. Figure 11A illustrates the five-level

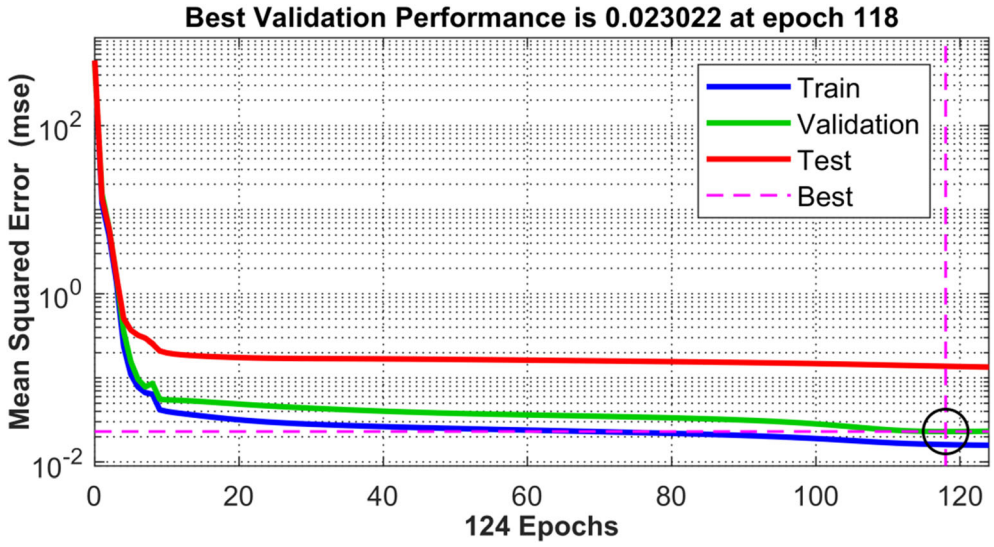


FIGURE 8 *ANN* convergence performance [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 2 Switching angles with respect to *M*, with ANN-based *SHE* control

		Switching angles	
Test	**Modulation index**	θ_1	θ_2
Scenario 1	None	30	60
Scenario 2	0.55	21	81
Scenario 3	0.65	11	71
Scenario 4	0.85	19	41

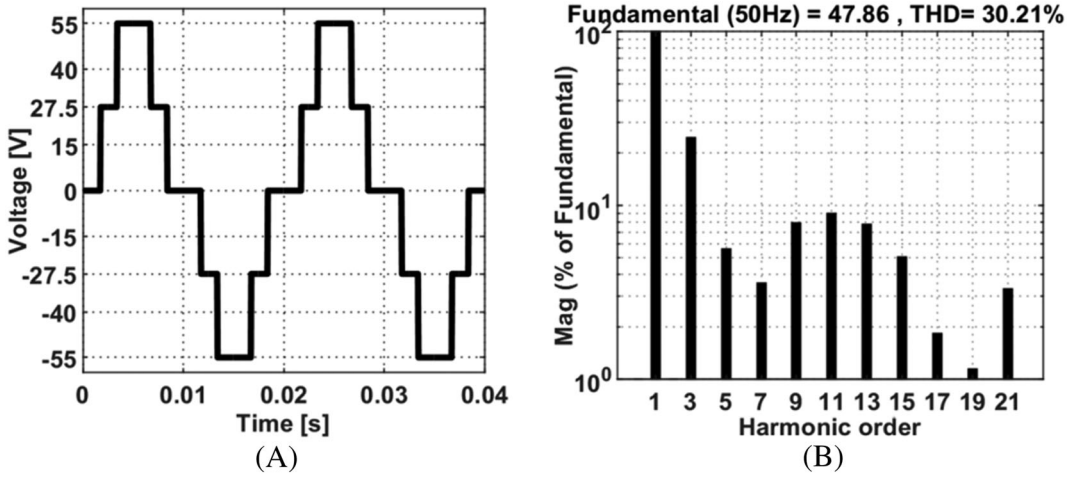


FIGURE 9 Voltage waveform and harmonic spectrum under Scenario 1: (A) The output voltage of the five-level inverter. (B) The associated harmonic spectrum

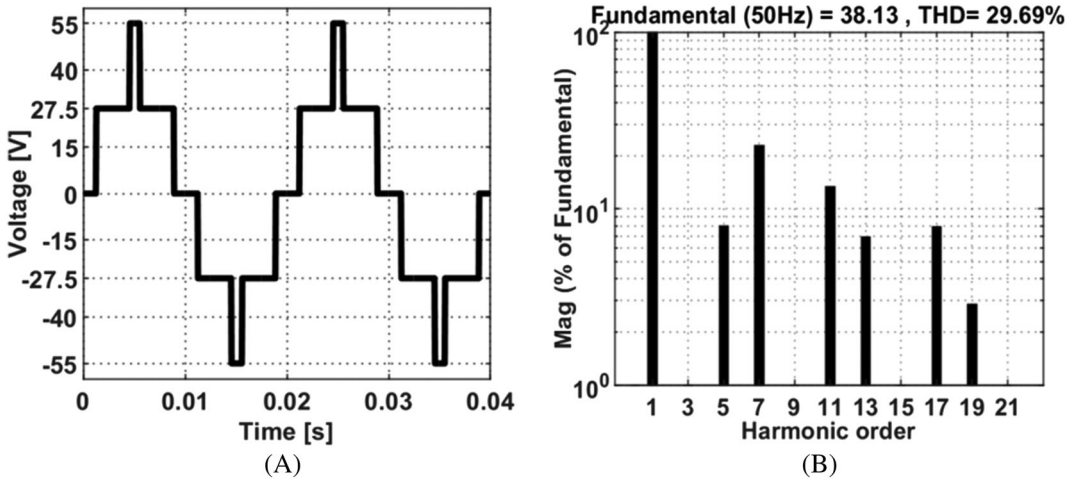


FIGURE 10 Voltage waveform and harmonic spectrum under Scenario 2: (A) The output voltage of the five-level inverter. (B) The associated harmonic spectrum

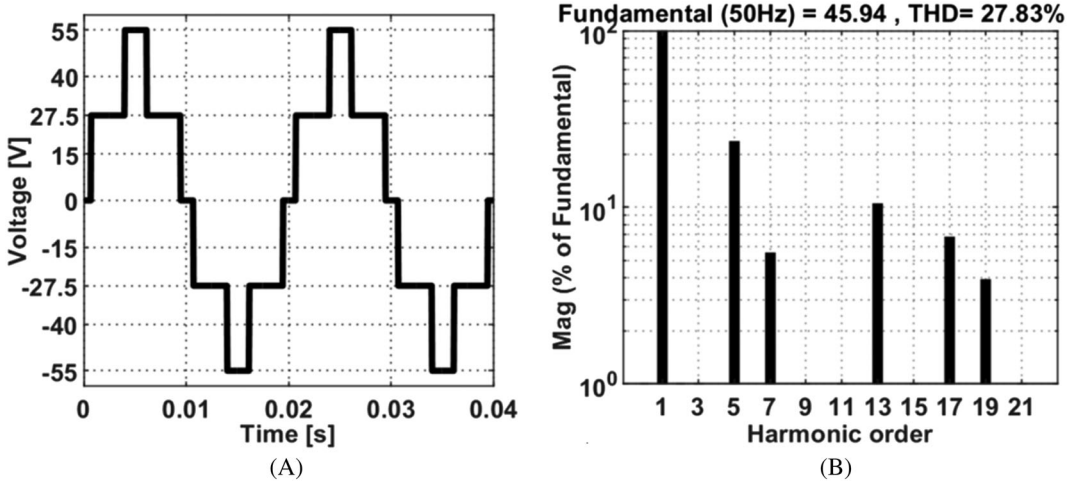


FIGURE 11 Voltage waveform and harmonic spectrum under Scenario 3: (A) The output voltage of the five-level inverter. (B) The associated harmonic spectrum

voltage output with *ANN-SHE* control under Scenario 3. Its associated harmonic spectrum is presented in Figure 11B. The *THD* voltage has a value of *(THDv* = 27.83%*)* and the third harmonic with their multiples have been eliminated. Figure 12A shows the five-level output voltage with *ANN-SHE* control under Scenario 4. Their associated harmonic spectrum is shown in Figure 12B. The *THDv* = 16.67% and the third harmonic with their multiples have been eliminated.

4.2 | Experimental results

The proposed *ANN-SHE* control is validated experimentally with a laboratory prototype of a five-level inverter built with Rohm (1200V-40A) MOSFETs and 02 *DC* power supplies. The control signals have been implemented in real-time

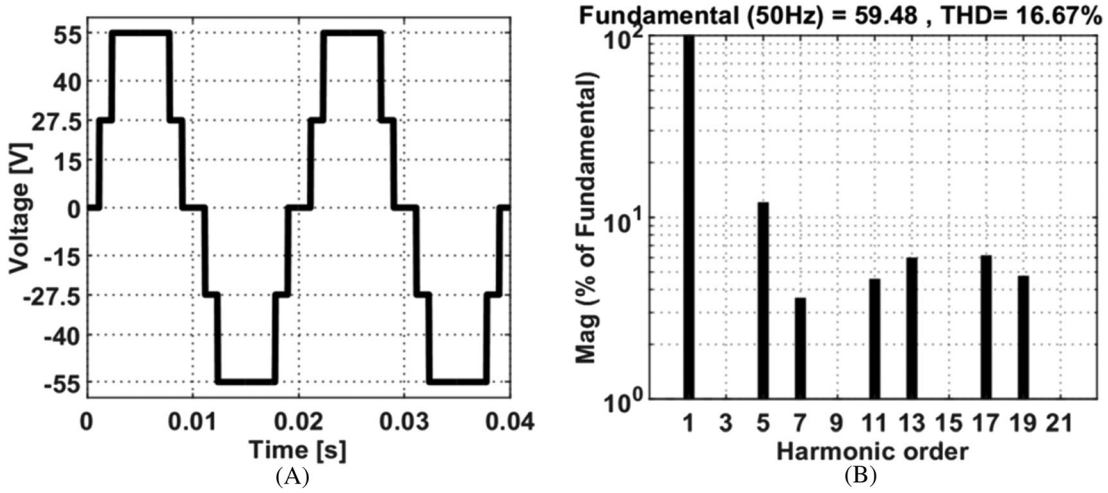


FIGURE 12 Voltage waveform and harmonic spectrum under Scenario 4: (A) The output voltage of the five-level inverter. (B) The associated harmonic spectrum

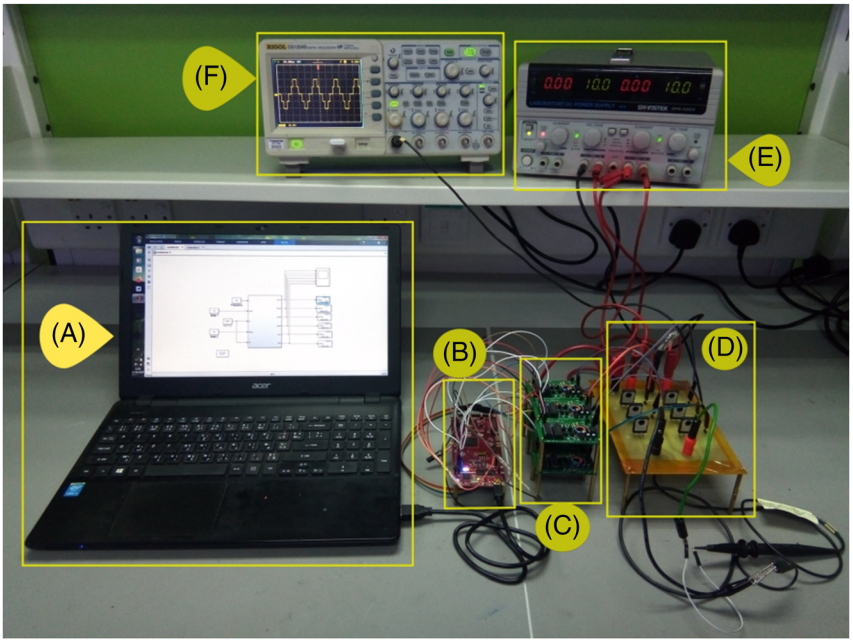


FIGURE 13 The experimental laboratory prototype. (A) Pc + Simulink C2000 package software; (B) C2000 Delfino–TMS320F28379D board; (C) Gate driver circuits; (D) MOSFET switches (inverter); (E) DC power supplies; (F) digital oscilloscope [Colour figure can be viewed at wileyonlinelibrary.com]

using LAUNCHXL-C2000 Delfino–TMS320F28379D board and PC equipped with Matlab/Simulink C2000 package software, as shown in Figure 13. The FLUKE 43B power quality analyzer is used to evaluate the harmonic spectrums and retrieve the values of *THDv*. Figure 14A shows the experimental gating signals S_1, S_3, and S_5. Figure 14B shows the experimental gating signals S_2, S_4, and S_6.

Figure 15A illustrates the experimental phase voltage of the five-level inverter without *SHE* control under Scenario 1. Figure 15B–D illustrates the experimental phase voltage of the five-level inverter with *ANN-SHE* control under Scenarios 2–4, respectively. Figure 16A–D illustrates the harmonics spectrum of the output five-level voltage under Scenarios 1–4, respectively. The obtained *THD* values of the output voltage have a value of 29.7%, 29.9%, 27.9%, and 17% under the four scenarios respectively. Furthermore, Figure 16B–D demonstrates that the third harmonic plus their multiples have been eliminated, and *THDv* can be minimized only to 17% in five-level inverter with the *SHE* method because *SHE* technique is low-frequency modulation technique. Also, maximum *RMS* voltage has been achieved with the appropriate switching angles, as shown in Figure 16D.

ANN-SHE control is a simple and easy technique that can be used with minimum knowledge required of applied mathematics compared to the conventional *SHE* technique, which needs an advanced level to solve the nonlinear equations. The obtained simulation and experimental results demonstrate the effectiveness of the proposed *ANN-SHE* control to reduce the *THDv* and to eliminate the third harmonic plus their multiples in the five-level inverter topology. A remarkable similarity exists between the simulation results and the experimental results, as presented in Table 3 and the presented histogram in Figure 17. High efficiency achieved with *THDv* of 17%. This value is outstanding for implementing a five-level inverter because the *SHE* technique is a low-frequency modulation technique. However, the

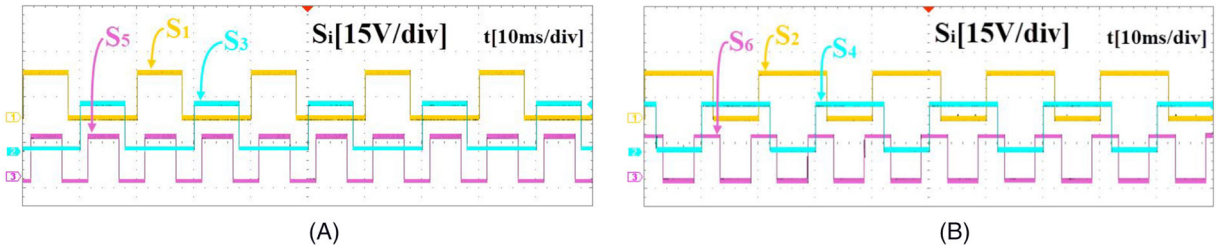


FIGURE 14 The experimental gate signals (A) of switches S_1, S_3, and S_5 and (B) of switches S_2, S_4, and S_6 [Colour figure can be viewed at wileyonlinelibrary.com]

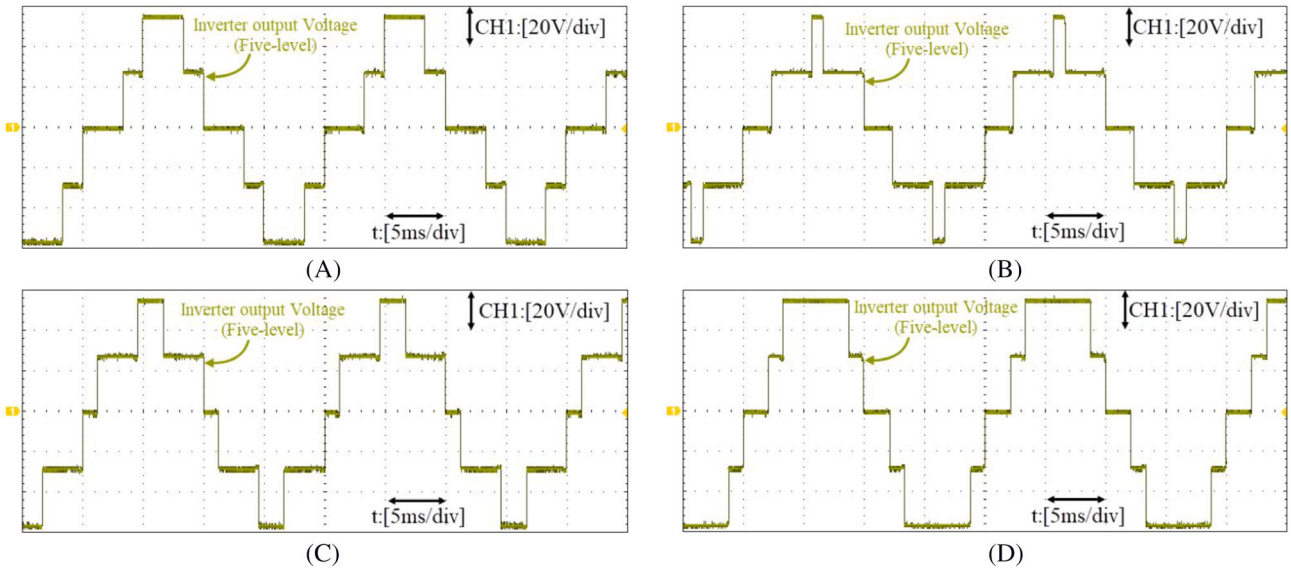


FIGURE 15 Experimental phase voltage of the five-level inverter with *ANN-SHE* control. (A) $\theta_1 = 30°$ and $\theta_2 = 60°$. (B) $M = 0.55$, $\theta_1 = 21°$ and $\theta_2 = 81°$. (C) $M = 0.65$, $\theta_1 = 11°$ and $\theta_2 = 71°$. (D) $M = 0.85$, $\theta_1 = 19°$ and $\theta_2 = 41°$ [Colour figure can be viewed at wileyonlinelibrary.com]

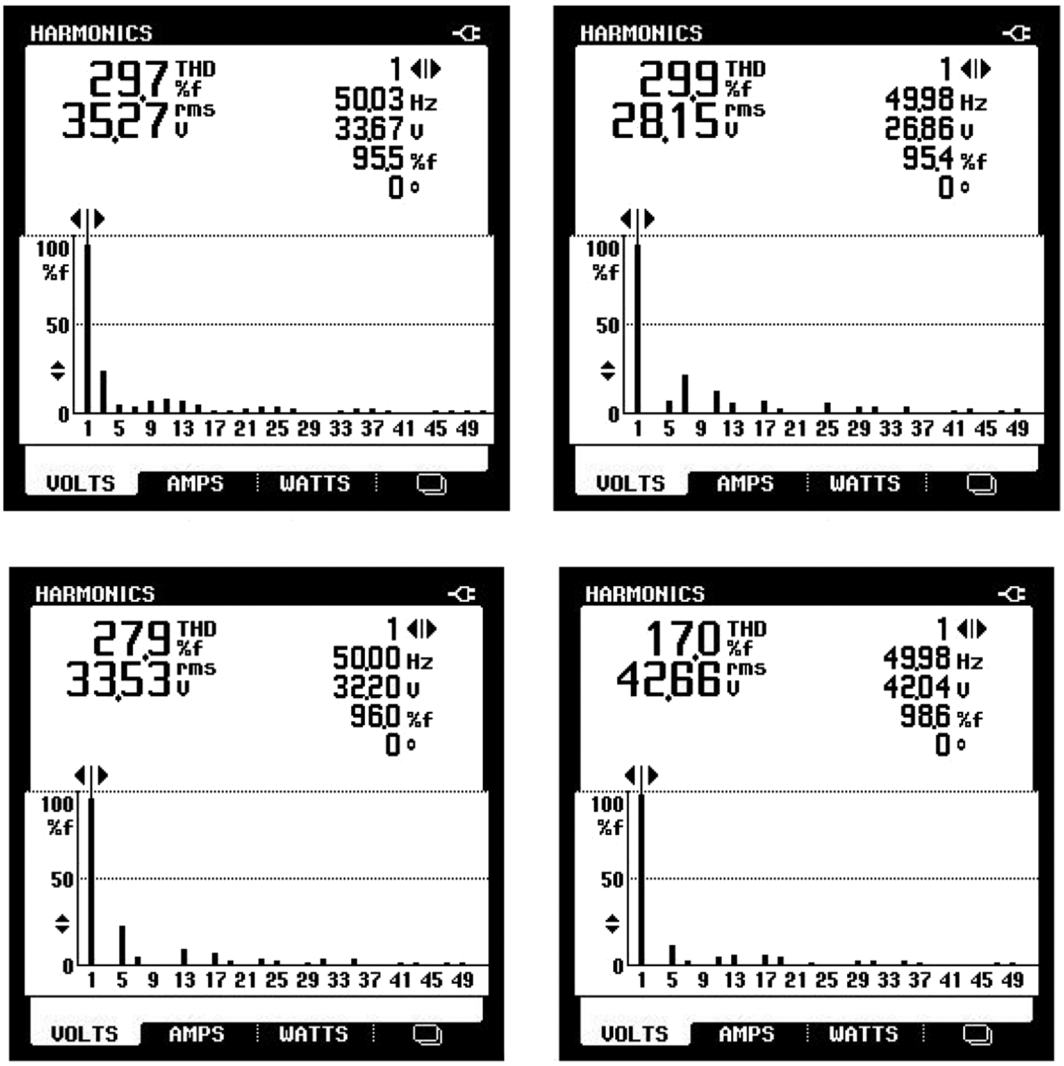


FIGURE 16 Experimental harmonics spectrum analysis of the output five-level voltage with (A) $\theta_1 = 30°$, $\theta_2 = 60°$. (B) $M = 0.55$, $\theta_1 = 21°$, $\theta_2 = 81°$. (C) $M = 0.65$, $\theta_1 = 11°$, $\theta_2 = 71°$. (D) $M = 0.85$, $\theta_1 = 19°$, $\theta_2 = 41°$

TABLE 3 Summarize the obtained results

Test	Simulation *THDv* (%)	Experimental *THDv* (%)
(1) $\theta_1 = 30°$, $\theta_2 = 60°$.	30.21	29.7
(2) $M = 0.55$, $\theta_1 = 21°$, $\theta_2 = 81°$.	29.69	29.9
(3) $M = 0.65$, $\theta_1 = 11°$, $\theta_2 = 71°$.	27.83	27.9
(4) $M = 0.85$, $\theta_1 = 19°$, $\theta_2 = 41°$.	16.67	17

THDv of the five-level inverter can be more improved using high-frequency modulation strategies as *SHEPWM* or using LC filter. But, the *SHEPWM* method requires few lines of code due to the numerous switching angles at each voltage level in the switching period (T), compared to *SHE*, where there are only two switching angles.

4.3 | Comparison of *ANN-SHE* technique with GA and N-R algorithms

The popular algorithms used to solve the *SHE* equations are numerical methods as *N-R*[25] and gradient optimization; algebraic methods as resultant theory and Wu method; intelligent techniques such as (*GA*), differential evolution (*DE*),

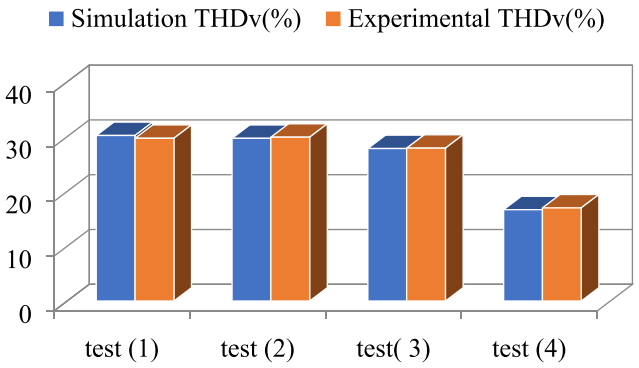


FIGURE 17 Histogram presentation of simulation and experimental *THDv* (%) versus various scenarios (test) [Colour figure can be viewed at wileyonlinelibrary.com]

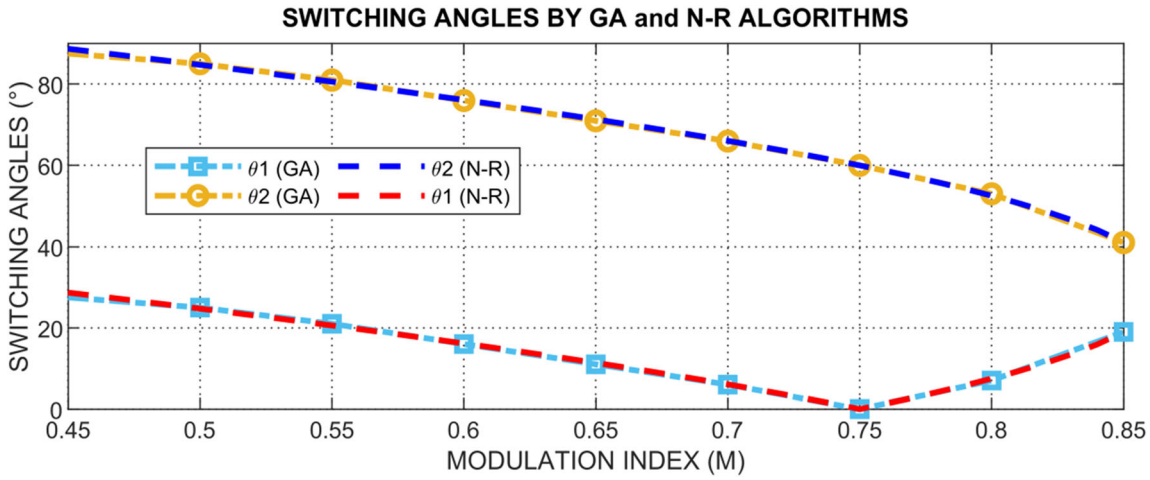


FIGURE 18 The switching angles θ_1 and θ_2 using *GA* and *N-R* algorithms [Colour figure can be viewed at wileyonlinelibrary.com]

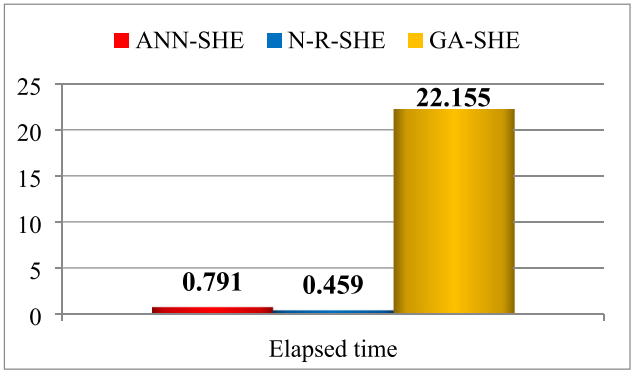


FIGURE 19 The execution time of the compared algorithms (seconds) [Colour figure can be viewed at wileyonlinelibrary.com]

and *PSO*.[26] In this subsection, the *SHE* equations are solved using two different algorithms, namely, *N-R* and *GA*. Figure 18 illustrates the achieved values of switching angles θ_1 and θ_2 with *GA* and *N-R* algorithms.

Figure 18 shows that the two algorithms give the same solutions (values of switching angles θ_1 and θ_2) as the *ANN-SHE* for the *M* range of 0.45–0.85. The three algorithms are attaining the objective function; for that, these algorithms can be classified just in terms of the complexity of implementation and the execution speed. Figure 19 shows the execution time required to compute the switching angles for each algorithm. The *ANN-SHE* requires

0.791 s, while the *N-R-SHE* requires only 0.459 s. The *GA-SHE* is very slow and requires 22.155 s. The *N-R* is the fastest algorithm because it is a numerical method. However, it requires a careful selection of the initial switching angles to ensure the convergence. This is the main drawback of the *N-R-SHE* algorithm. There is no systematic way to select the initial solution. Several tests are required to get the *N-R* algorithm converges to the solution. The *GA* algorithm is an easy method, where the solutions of *SHE* equations are found in configured space and constraints. But this algorithm needs a few seconds in order to calculate the switching angles which make it impractical for the real-time applications. The *ANN* is the best solution for solving the SHE equations compared to *GA* and *N-R* methods because it has a very low computational burden; it requires less than 1 s to provide the solution. Furthermore, it does not require any initialization or parameter tuning to converge to the optimal solution, thus makes it suitable for real-time applications.

4.4 | Comparison of *ANN-SHE* with other switching modulation techniques

The five-level inverter can be controlled using various modulation strategies. To presents the superiority of the proposed low-frequency *ANN-SHE* technique, a comparison between a high-frequency switching modulation (*PD-LP-PWM*), a low-frequency switching half-height (*HH*) method,[27] a low-frequency switching equal-phase (*EP*) method,[27] and the discussed *ANN-SHE* method is carried out in this subsection. Figure 20 illustrates the principle of level-shifted (*LS-PWM*) modulation based on phase disposition (*PD*), and the waveform of the five-level output voltage. The principle of *PD-LP-PWM* technique is to compare triangular carriers with the sinusoidal reference waveform to produce the gating signals of switches. Figure 21 shows *THD* comparison using some modulations strategies with dynamic variation of the modulation index *M*.

From Figure 21, it is shown that the low-frequency category (*SHE* with *ANN*, *GA*, and *N-R*) algorithms have a better *THD* response over *PD-LS-PWM* and *EP* method. Also, *SHE* has a better *THD* response over *HH* method for a modulation value of 0.85. The *HH* method has a better *THD* response compared to the *SHE* in *M* range of 0.5–0.8, but the switching angles with *EP* method and *HH* method do not depend on modulation index *M*. Table 4 summarizes the modulation strategies with their best achievable *THD* of each one of them.

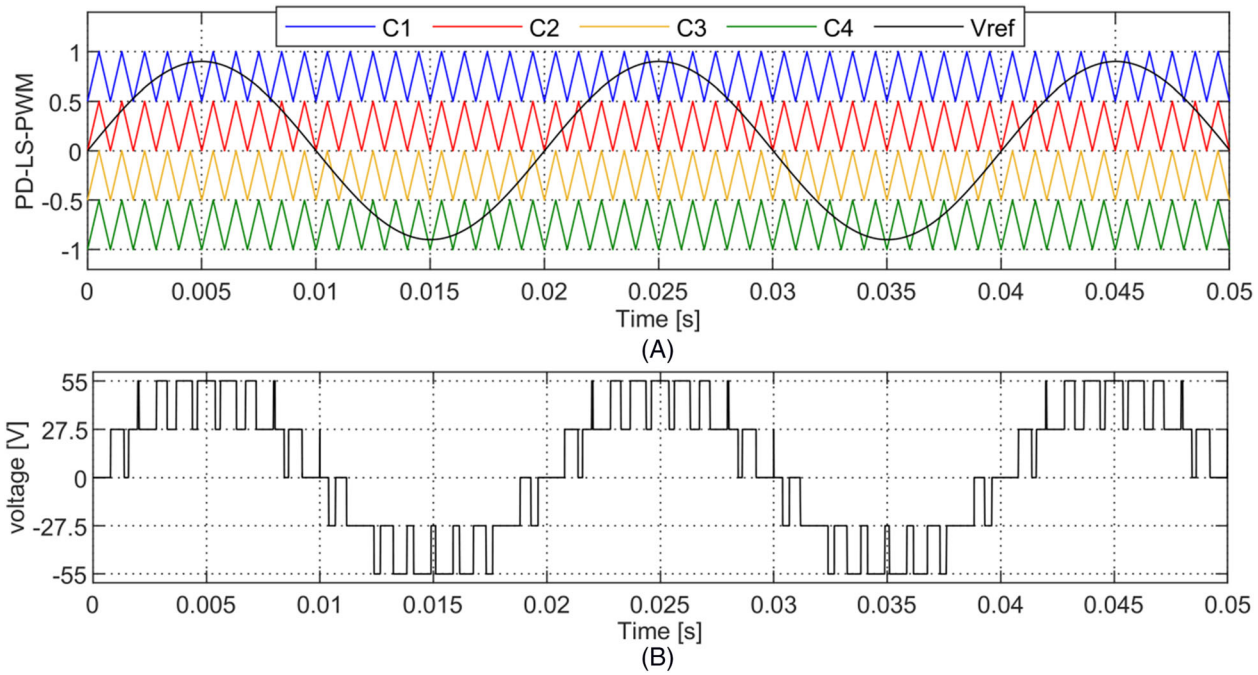


FIGURE 20 Multicarrier *PWM* modulation technique for the five-level inverter. (A) The level-shifted (*LS-PWM*) modulation based on phase disposition (*PD*). (B) The waveform of the five-level output voltage using (*PD-LS-PWM*) [Colour figure can be viewed at wileyonlinelibrary.com]

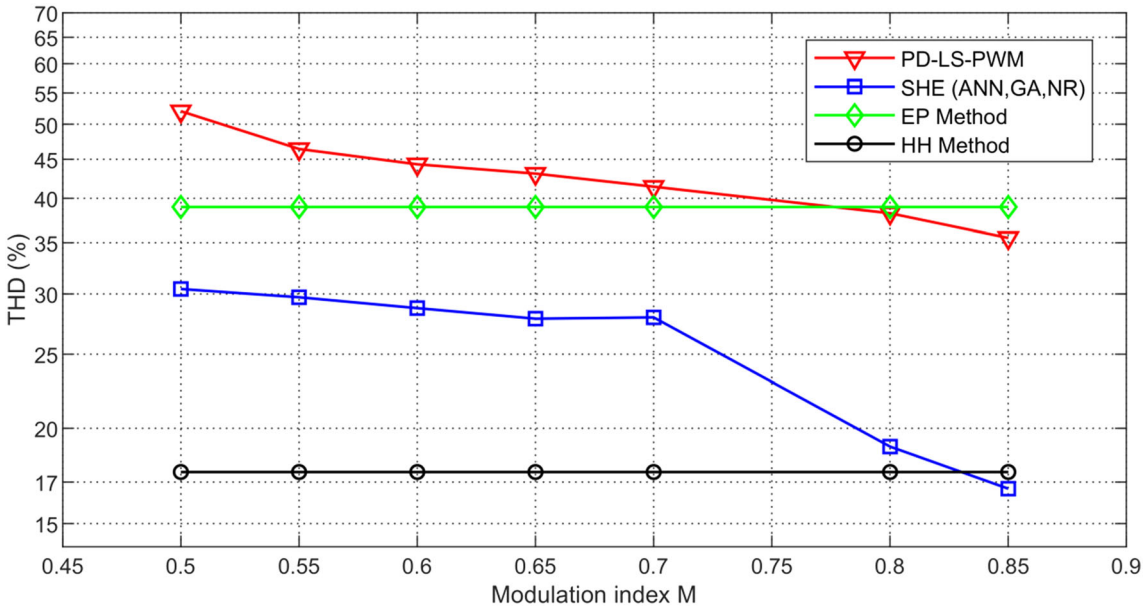


FIGURE 21 Comparison of some modulation strategies in terms of *THD* [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 4 Summarize the low *THD*

Modulation strategy	Switching strategy	Low *THD* (%)
EP method[27]	Low-frequency	38.96
HH method[27]	Low-frequency	17.53
PD-LS-PWM	High-frequency	35.51
NR-SHE	Low-frequency	17
GA-SHE	Low-frequency	17
ANN-SHE	Low-frequency	17

5 | APPLICATION AREAS OF THE FIVE-LEVEL INVERTER WITH *ANN-SHE*

The presented five-level inverter with *ANN-SHE* is suitable for various applications in power electronic systems and industrial process. The most important applications including the resonant inverters and grid-connected systems. A short review of the inverter applications is given in this section.

5.1 | Grid-connected systems

One of the possible implementations of the proposed *ANN-SHE* controller with the five-level inverter topology is to be used for grid-connection through an R-L, LCL, or LC filters. These filters are the most popular and often used to minimize the *THD* according the international standards. The *SHE* plays an important role for reduction of filter size during the design stage of the system. Figure 22 illustrates the circuit of the grid-connected five-level inverter through an LCL filter.

5.2 | Five-level resonant inverters

The presented structure of a five-level inverter can be used to construct resonant inverters.[28] The resonant inverters are particular converters operated with high-frequency values, generally superior to 1 kHz, to produce a high-frequency

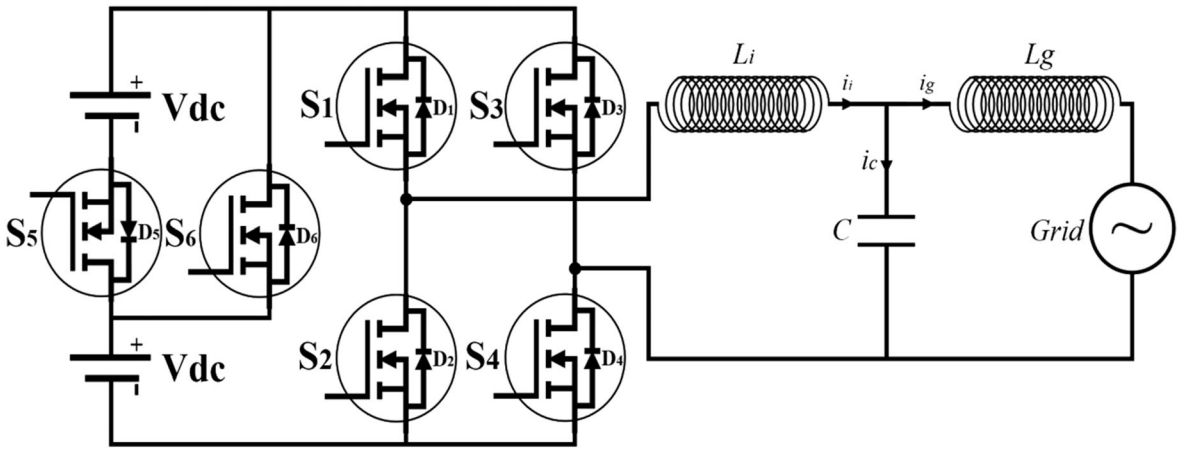


FIGURE 22 Grid-connected five-level inverter through an LCL filter

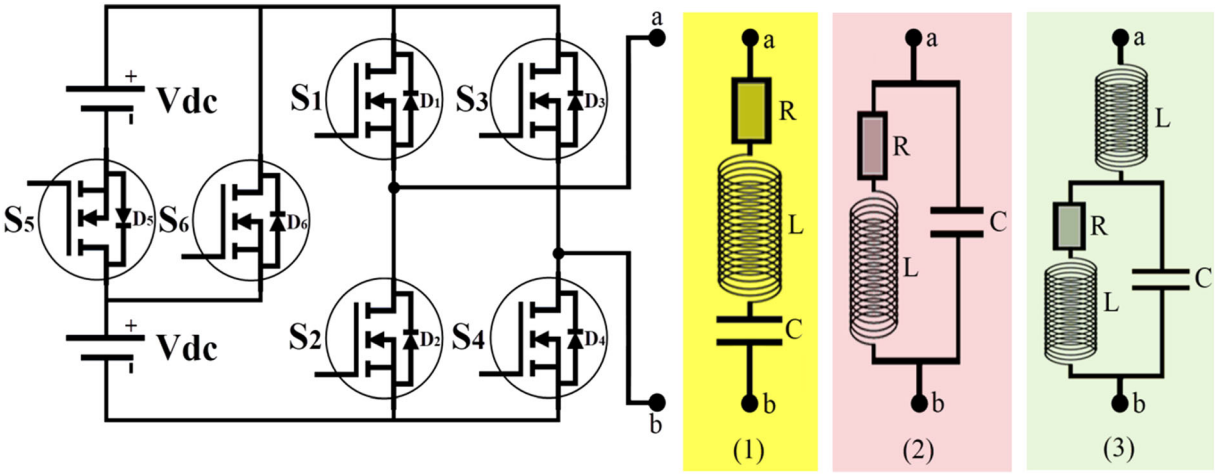


FIGURE 23 The five-level resonant inverter configurations [Colour figure can be viewed at wileyonlinelibrary.com]

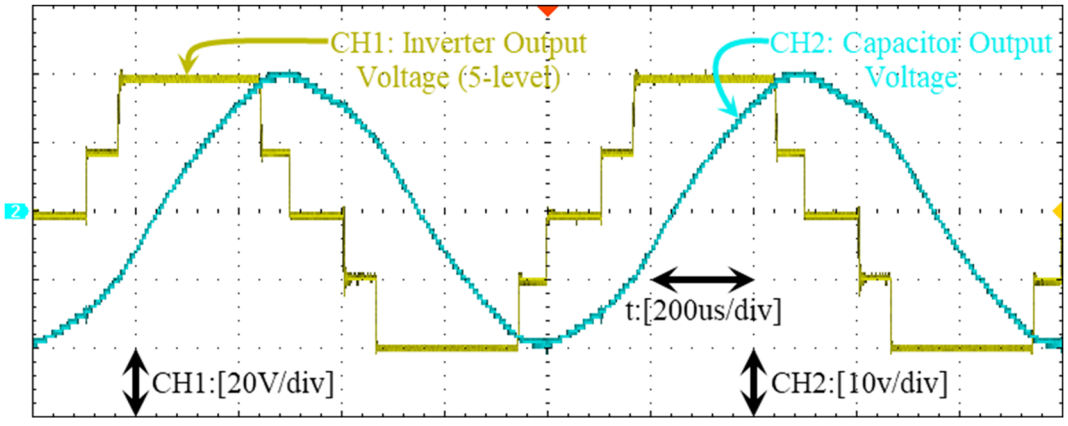


FIGURE 24 The essential waveforms of the series resonant inverter with $f = 1$ kHz, $\theta_1 = 19°$, $\theta_2 = 41°$ [Colour figure can be viewed at wileyonlinelibrary.com]

output voltage and current. Resonant inverter is characterized by high efficiency, high-power factor mode, and low harmonic currents.[29,30] Also, it is used in several industrial applications including: the induction heating, the wireless power transfer, and the telecommunications.[31] Figure 23 illustrates the three types of resonant inverters: (1) the series, (2) the parallel, and (3) the series parallel. Figure 24 illustrates the essential waveforms of the series resonant inverter

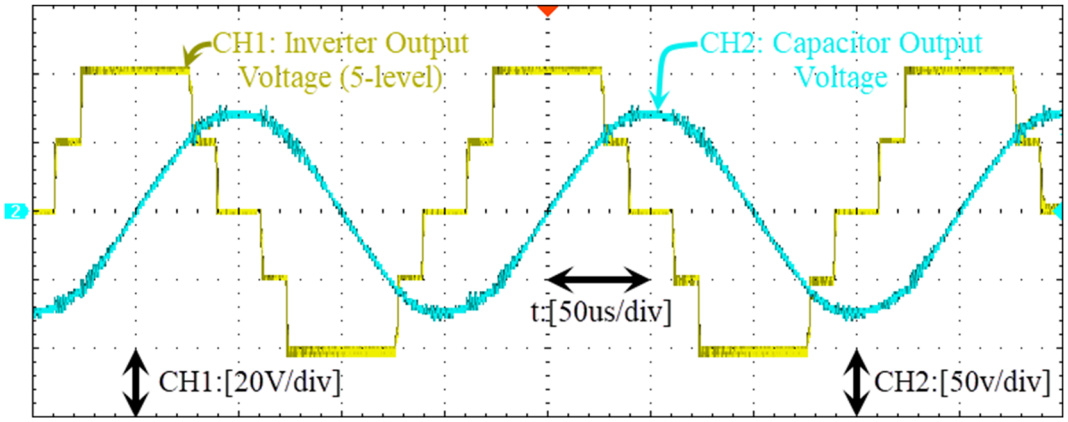


FIGURE 25 The essential waveforms of the series resonant inverter with $f = 5$ kHz, $\theta_1 = 19°$, $\theta_2 = 41°$ [Colour figure can be viewed at wileyonlinelibrary.com]

with $f = 1$ kHz, $\theta_1 = 19°$, $\theta_2 = 41°$. Figure 25 illustrates the essential waveforms of the series resonant inverter with $f = 5$ kHz, $\theta_1 = 19°$, $\theta_2 = 41°$.

5.3 | Energy and power systems

The proposed five-level inverter with *ANN-SHE* controller is suitable for renewable energy and standalone systems, where the produced energy in *DC* form needs to be converted to *AC* form because most electrical equipment works with alternative currents. The five-level inverter is able to adjust the motor speed by controlling the frequency of the alternating current. So, the inverter is suitable for transportation electrification applications. Also, the inverter is an essential element in the distributed generation systems (*DGS*), where *DGS* have been attracted a lot of attention of researchers because their advantages, such as stability of electrical energy production, fewer power losses, and less cost compared to centralized generation systems.[32]

6 | CONCLUSION

This paper aimed to demonstrate the use of a *NN* method to control a hybrid five-level inverter with the *ANN-SHE* technique. The simulation was carried out under Matlab/Simulink environment to confirm the performance of the proposed strategy. Furthermore, an experimental validation has been performed using a low-cost *DSP* C2000 Delfino-28379D microcontroller embedded board. Experimental results demonstrate that the proposed *ANN-SHE* control of *MLI* is an efficient and straightforward technique. The selected harmonics have been eliminated where a *THDv* of 17% was achieved with the five-level inverter. The achieved *THDv* value was outstanding for five-level inverter because the *SHE* technique is a low-frequency modulation technique. Notwithstanding these advantages, *THDv* can be more improved using a high-frequency modulation strategy (*SHEPWM*). Surprisingly, this is easy and possible because its implementation needs just a few extra lines of code. This paper is expected to serve as a reference for future studies of using the *ANN* for the control of electrical systems.

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CONFLICT OF INTEREST

There is no conflict of interest to declare.

DATA AVAILABILITY STATEMENT

Research data are not shared.

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